

Supplementary Online Content

Stunnenberg BC, Raaphorst J, Groenewoud HM, et al. Effect of mexiletine on muscle stiffness in patients with nondystrophic myotonia evaluated using aggregated N-of-1 trials. *JAMA*. doi:10.1001/jama.2018.18020

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This supplementary material was provided by the authors to give readers additional information about their work.

eTable 1. Inclusion and Exclusion Criteria

Inclusion criteria	Exclusion criteria
1. At least 18 years of age	1. Inability or willingness to approved to provide informed consent
2. Genetically confirmed diagnosis of NDMs	2. Other neurological conditions that might affect the assessment of the study measurement
	3. Genetically confirmed DM1 (CTG > repeats), or DM2
	4. Existing cardiac conduction defects, evidenced on ECG including but not limited to the following condition: malignant arrhythmia or cardiac conduction disturbance (such as second degree AV block, third degree AV block, or prolonged QT interval >500 ms or QRS duration > 150 msec)
	5. Current use of the following antiarrhythmic medication for a cardiac disorder: flecainide acetate, encainide, disopyramide, procainamide, quinidine, propafenone or mexiletine
	6. Women who are pregnant or lactating
	7. Currently on medication for myotonia such as phenytoin and flecainide acetate within 5 days of enrollment, carbamazepine and mexiletine within 3 days of enrollment, or propafenone, procainamide, disopyramide, quinidine and encainide within 2 days of enrollment
	8. Renal or hepatic disease, heart failure, history of myocardial infarction, or seizure disorders

Abbreviations: NDMs= non-dystrophic myotonic syndromes; DM= myotonic dystrophy; CTG= cysteine-threonine- glycine (amino acids); AV= atrioventricular. This table was reproduced with permission of BioMed Central as the original publisher from the published study protocol (*Stunnenberg BC, Woertman W, Raaphorst J, et al. Combined N-of-1 trials to investigate mexiletine in non-dystrophic myotonia using a Bayesian approach; study rationale and protocol. BMC neurology 2015;15:43*).

eTable 2. Individual Demographic, Genotype and IVR-Baseline Information per Study Patient

Patient nr.	Age (years)	Gender	Affected gene	Mutation(s) (allele 1/ allele 2)	Previous anti-myotonic treatment	Relevant comorbidity	IVR – stiffness (mean)	IVR – pain (mean)	IVR – weakness (mean)	IVR – tiredness (mean)
1.	24	Male	<i>CLCN1</i>	Gly233Ser / Gly233Ser	Mexiletine, acetazolamide	-	7.87	1.20	9.00	8.80
2.	33	Male	<i>CLCN1</i>	Gly285Glu / Arg894X	Phenytoin	Asthma	6.93	0.00	6.00	2.93
3.	27	Male	<i>CLCN1</i>	c.302-1G>A	-	-	2.85	2.31	0.00	0.00
4.	25	Female	<i>SCN4A</i>	Gly1306Ala	-	-	4.93	2.93	2.36	2.57
5.	25	Male	<i>CLCN1</i>	Arg894X / Arg894X	Phenytoin	-	7.07	0.00	0.00	2.36
6.	26	Male	<i>CLCN1</i>	Gly233Ser / Gly233Ser	Acetazolamide, carbamazepine	Cluster headache	7.78	6.11	7.89	7.44
7.	39	Female	<i>SCN4A</i>	Gly305Glu / Ser264fsX6	-	Depression	5.20	0.00	0.00	4.47
8. *	21	Female	<i>SCN4A</i>	Gly1306Val	Acetazolamide	-	5.14	3.68	0.00	5.41
9. *	51	Male	<i>SCN4A</i>	Gly1306Val	Acetazolamide, gabapentin	-	5.09	5.55	0.00	6.14
10.	32	Male	<i>SCN4A</i>	Gly1306Ala	-	Panic disorder	4.71	2.29	0.00	7.29
11.	53	Male	<i>SCN4A</i>	Gly1306Val	Amitriptyline	Stable angina pectoris	0.00	5.12	0.00	4.47
12.	53	Female	<i>SCN4A</i>	Val1485Phe	-	Migraine, artrosis, asthma	3.56	3.37	3.75	4.88
13.	51	Male	<i>SCN4A</i>	Arg1148Cys	-	Lumbago	4.23	5.46	4.38	5.46
14. **	54	Female	<i>CLCN1</i>	c.302-2 A>C	-	-	5.50	0.00	3.00	5.08
15. **	58	Male	<i>CLCN1</i>	c.302-2 A>C	-	Hypertension	7.07	0.47	3.67	2.60
16. **	60	Male	<i>CLCN1</i>	c.302-2 A>C / Met646Ile	Quinidine	-	6.62	0.00	4.00	6.31
17.	55	Male	<i>CLCN1</i>	Gly285Glu / Met646Ile	-	Hernia lumbosacralis	5.87	3.67	5.13	1.07
18.	48	Female	<i>CICN1</i>	c.302-2 A>C / Met646Ile	Tocainide	-	5.71	0.59	4.18	0.88
19. *	21	Male	<i>SCN4A</i>	Gly1306Val	-	-	3.31	2.00	0.23	3.85

20.	32	Female	SCN4A	Gly1306Val	-	-	6.08	5.85	0.46	5.08
21.	64	Male	SCN4A	Gly1306Val	Acetazolamide	COPD	2.77	3.00	2.62	2.62
22.	19	Male	SCN4A	Leu703Pro	Acetazolamide	-	8.00	0.00	5.00	7.00
23.	50	Male	CLCN1	Gly285Glu / Met646Ile	Acetazolamide	-	7.71	0.00	7.71	0.00
24. ***	53	Male	CLCN1	Phe413Cys	-	-	6.80	3.73	4.60	7.47
25. ***	58	Male	CLCN1	Phe413Cys	-	Hypothyroidism	6.57	0.00	2.79	3.07
26. ***	62	Male	CLCN1	Phe413Cys	-	-	6.80	0.00	3.93	3.67
27. ***	60	Male	CLCN1	Phe413Cys	-	Cerebral infarction	6.22	0.00	0.00	6.89
28.	65	Male	CLCN1	Lys195fsX	Quinidine	Subdural hematoma	6.71	0.00	3.37	0.79
29.	44	Female	CLCN1	Phe413Cys	-	-	6.71	0.00	0.57	4.86
30.	49	Male	CLCN1	Gly285Glu	-	Panic disorder	4.62	3.69	0.00	4.00

*/**/*** Study patients belonging to the same family. Abbreviations: IVR= interactive voice response system; *CLCN1*=skeletal muscle chloride channel gene; *SCN4A*= skeletal muscle sodium channel gene.

eTable 3. Results of Individual Bayesian Interim Analyses on the Primary Outcome Measure

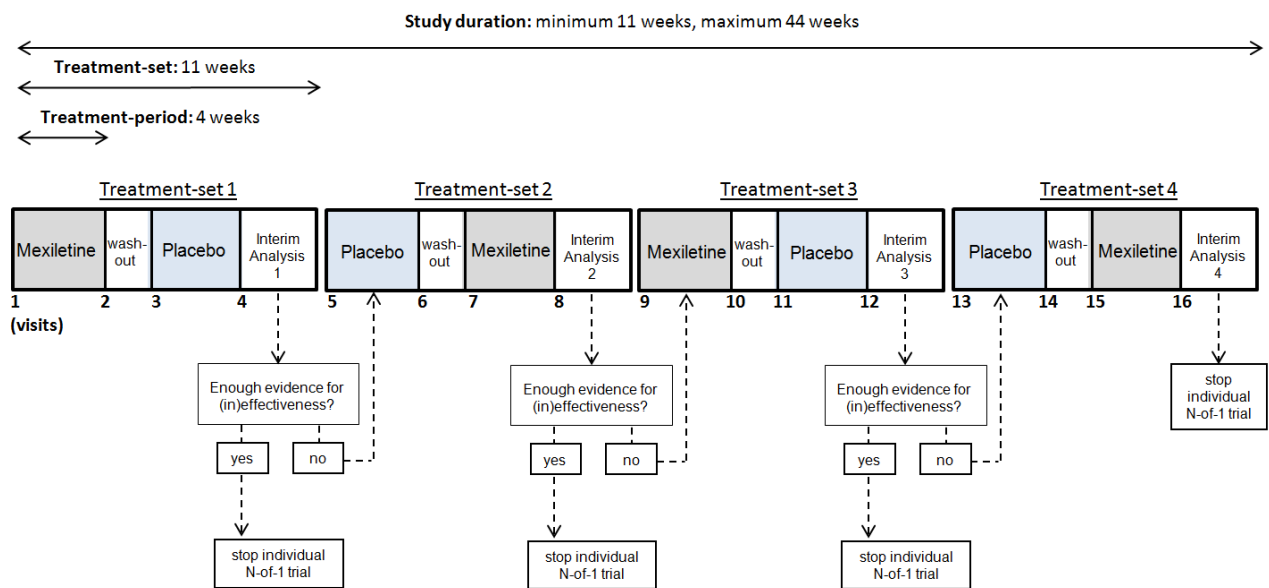
Patient nr.	Affected gene	Interim analysis nr.	Mean treatment effect	95% Credible Interval	Posterior probability (of clinically meaningful difference)	Mexiletine treatment post-trial
1	<i>CLCN1</i>	2	4.4	(4.03 ; 4.85)	100%	Continued
2	<i>CLCN1</i>	2	3.1	(2.63 ; 3.49)	100%	Continued
4	<i>SCN4A</i>	2	1.1	(0.27 ; 1.90)	81%	Continued
5	<i>CLCN1</i>	2	4.7	(4.50 ; 4.87)	100%	Continued
8	<i>SCN4A</i>	1	1.2	(-0.03 ; 2.41)	80%	Continued
9	<i>SCN4A</i>	1	3.2	(2.78 ; 3.56)	100%	Continued
10	<i>SCN4A</i>	1	3.7	(2.28 ; 5.04)	100%	Continued
11	<i>SCN4A</i>	1	0.00	(-12.24 ; 12.16)	38%*	Stopped
12	<i>SCN4A</i>	1	2.66	(1.76 ; 3.56)	100%	Continued
13	<i>SCN4A</i>	1	-0.05	(-0.56 ; 0.45)	2%	Stopped
14	<i>CLCN1</i>	1	3.5	(2.81 ; 4.19)	100%	Continued
15	<i>CLCN1</i>	1	2.7	(2.16 ; 3.24)	100%	Continued
16	<i>CLCN1</i>	1	4.49	(3.96 ; 5.03)	100%	Continued
17	<i>CLCN1</i>	1	3.22	(2.94 ; 3.50)	100%	Continued
18	<i>CLCN1</i>	1	2.55	(1.77 ; 3.32)	100%	Continued
19	<i>SCN4A</i>	1	2.63	(1.18 ; 4.01)	99%	Continued
20	<i>SCN4A</i>	1	1.13	(0.14 ; 2.13)	80%	Continued
21	<i>SCN4A</i>	1	4.41	(3.80 ; 5.0)	100%	Continued
22	<i>SCN4A</i>	1	0.01	(-0.56 ; 0.57)	1%	Stopped
23	<i>CLCN1</i>	1	5.49	(4.98 ; 6.04)	100%	Continued
24	<i>CLCN1</i>	1	2.64	(2.16 ; 3.12)	100%	Continued
25	<i>CLCN1</i>	1	1.65	(1.27 ; 2.03)	100%	Continued
26	<i>CLCN1</i>	1	4.65	(4.07 ; 5.21)	100%	Continued
27	<i>CLCN1</i>	1	3.14	(2.46 ; 3.83)	100%	Continued
28	<i>CLCN1</i>	1	4.50	(4.0 ; 4.99)	100%	Continued
29	<i>CLCN1</i>	1	6.99	(6.59 ; 7.41)	100%	Continued
30	<i>CLCN1</i>	1	3.57	(2.91 ; 4.23)	100%	Continued

Primary outcome measure (IVR-stiffness) mean treatment effects are reported (N=27). Mexiletine non-responders (pre-defined posterior probability effect of reaching 0.75 point difference on the IVR-scale <0.20, i.e. 20%) are marked in bold (patient nr. 11, 13 and 22). Mexiletine-responders were pre-defined as posterior probability effect 0.75 point difference on the IVR-scale >0.80, i.e. 80%. Patient no. 1, 2, 4 and 5 represent patients that had two instead of one treatment-set as part of their individual N-of-1 trial. Patient no. 3, 6 and 7 dropped out. * Posterior probability in patient no. 11 is strongly influenced by the flat (wide) non-informative prior since the distribution of the treatment effect was 0 (95% CI 0;0) (absence of muscle stiffness during both placebo en mexiletine treatment periods). Therefore the individual N-of-1 trial was stopped after 1 treatment-set and mexiletine treatment was discontinued.

eTable 4. Adverse Reactions

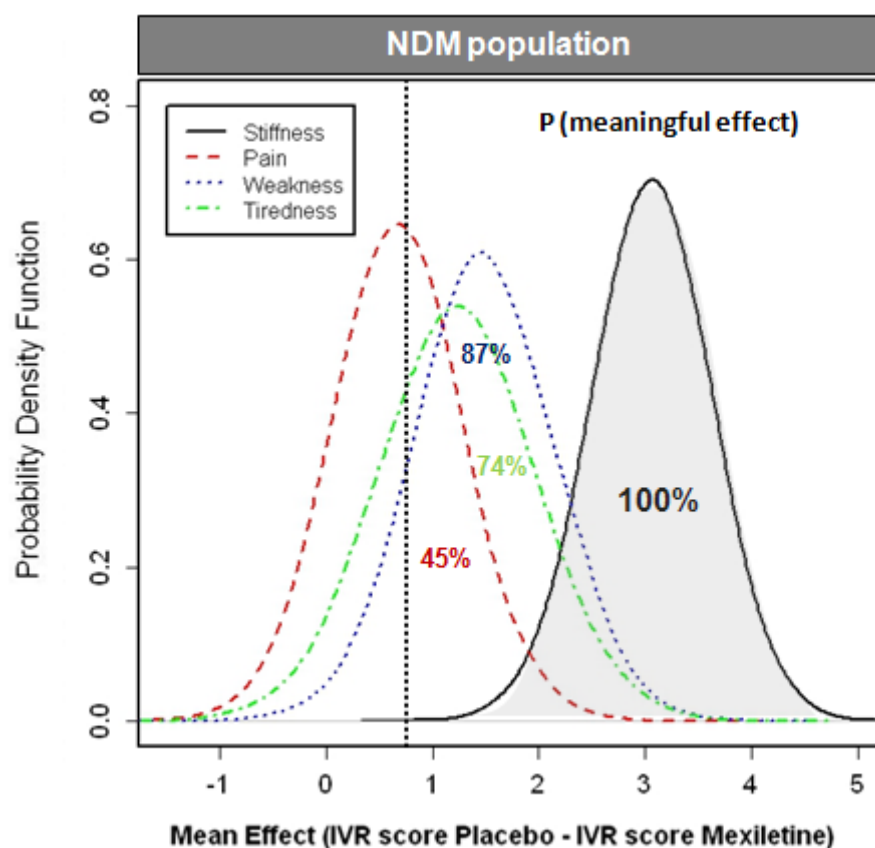
Adverse reactions	Mexiletine No. (%)	Placebo No. (%)
Gastro-intestinal discomfort (Reflux, dyspepsia, nausea, diarrhea and flatulence)	21 (70%)	1 (3%)
Tremor	2 (7%)	0 (0%)
Palpitations	2 (7%)	0 (0%)
Headache	1 (3%)	1 (3%)
Insomnia	1 (3%)	0 (0%)
Any	27 (90%)	2 (6%)
>1 adverse reactions	2 (7%)	0 (0%)
>2 adverse reactions	0 (0%)	0 (0%)
Causing study withdrawal	1 (3%)	0 (0%)
Leading to dosage reduction	3 (10%)	0 (0%)
Leading to additional therapy	2 (7%)	0 (0%)
Serious adverse reactions	Mexiletine No. (%)	Placebo No. (%)
Allergic skin reaction (toxicodermia)	1 (3%)	0 (0%)

eFigure 1. Study Design



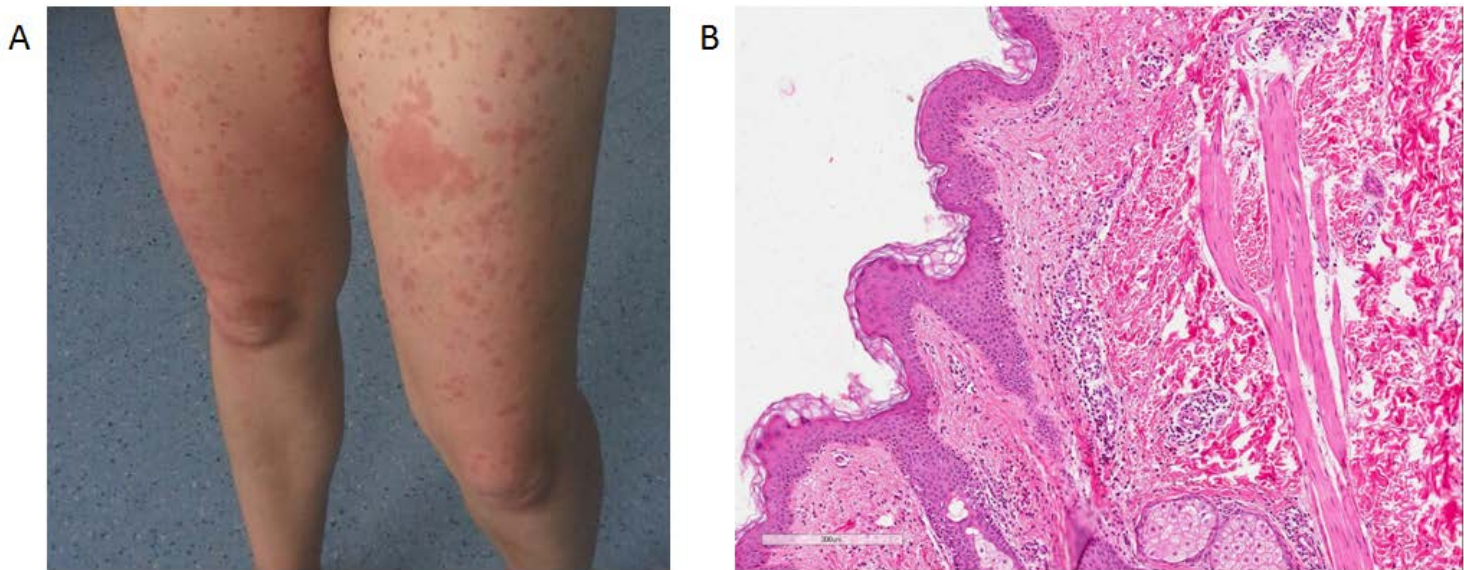
Each individual N-of-1 trial had a flexible design of a minimum of one and a maximum of four treatment-sets, depending of the level of evidence that was reached after each individual interim analysis at the end of each treatment-set. Note that in the trial, only four patients (no. 1, 2, 4 and 5) needed two treatment-sets to reach the predefined evidence for (in)effectiveness, while all other 23 patients needed only one treatment-set. This figure was reproduced with permission of BioMed Central as the original publisher from the published study protocol (*Stunnenberg BC, Woertman W, Raaphorst J, et al. Combined N-of-1 trials to investigate mexiletine in non-dystrophic myotonia using a Bayesian approach; study rationale and protocol. BMC neurology 2015;15:43*).

eFigure 2. Posterior Probability Distributions on Reaching a Clinically Meaningful Treatment Effect on NDM Disease Group Level (N=27)



The probability density functions (PDF) of the treatment effects of the four IVR-scores show the distributions around each mean treatment effect. The dotted line (0.75) represent the threshold for a clinically meaningful treatment effect. Posterior probabilities of a clinically meaningful treatment effect represent the area under the curve (AUC) of the treatment effect distributions right from the dotted line; p (meaningful effect) = 100% for IVR-stiffness, 45% for IVR-pain, 87% for IVR-weakness and 74% for IVR-tiredness.

eFigure 3. Serious Adverse Event in Patient No. 7



(A) Generalised erythematous papules that developed in patient no. 7 after eleven days of mexiletine treatment. (B) Histological section of skin biopsy shows a superficial dermatitis with eosinophilic infiltration and leucostasis as can be seen in toxicoderma (hematoxylin-eosin 40x).

eMethods 1. Winbugs Bayesian Model to Estimate the Effect for an Individual Patient During Interim Analyses

```
model
{
  for (i in 1:30) {
    Stiffness_plac[i]~dnorm(mu_plac,tau_plac)}
  for (j in 1:30) {
    Stiffness_mex[j]~dnorm(mu_mex,tau_mex)}
  mu_plac~dnorm(0, 0.001)
  mu_mex~dnorm(0,0.001)
  tau_plac<-1/(sigma_plac*sigma_plac)
  sigma_plac~dunif(0.01, 10)
  tau_mex<-1/(sigma_mex*sigma_mex)
  sigma_mex~dunif(0.01, 10)
  diff<-mu_plac-mu_mex
  p1<-step(diff)
  p2<-step(diff-0.75)
  p3<-step(diff-2)
}
```

In which:Real data:

Stiffness_plac[i] - IVR score of measurement i during placebo treatment

Stiffness_mex[i] - IVR score of measurement i during mexiletine treatment

Estimated model parameters:

<i>mu_plac</i>	Mean IVR score over the days with placebo treatment
<i>sigma_plac</i>	Standard deviation of IVR scores over the days with placebo treatment
<i>tau_plac</i>	$1/(\text{sigma_plac} * \text{sigma_plac})$
<i>mu_mex</i>	Mean IVR score over the days with mexiletine treatment
<i>sigma_mex</i>	Standard deviation of IVR scores over the days with mexiletine treatment
<i>tau_mex</i>	$1/(\text{sigma_mex} * \text{sigma_mex})$
<i>diff</i>	$\text{mu_plac} - \text{mu_mex}$ Difference between mean IVR score during placebo treatment en mean IVR score during mexiletine treatment
<i>p1</i>	Probability that the difference will be greater than 0.
<i>p2</i>	Probability that the difference will be greater than 0.75
<i>p3</i>	Probability that the difference will be greater than 2

In the model it is assumed there is a maximum number of 30 measurements during a treatment period.

eMethods 2. Winbugs Bayesian Hierarchical Model to Estimate the Mean Effect of Mexiletine in Total NDM Disease Group

```
model
{
  for( i in 1 : n ) {
    mu_mex[i] ~ dnorm(mu.mex, tau.mex)
    mu_plac[i] ~ dnorm(mu.plac, tau.plac)
    diff_patient[i] <- mu_plac[i] - mu_mex[i]
    for ( t in 1 : N_plac[i] ) {
      Stiff_Plac[i, t] ~ dnorm(mu_plac[i], tau.within)
    }
    for ( t in 1 : N_mex[i] ) {
      Stiff_Mex[i, t] ~ dnorm(mu_mex[i], tau.within)
    }
  }
  p1 <- step(mu.plac - mu.mex)
  p2 <- step(mu.plac - mu.mex - 0.75)
  p3 <- step(mu.plac - mu.mex - 2)
  tau.within<-1/(sigma.within*sigma.within)
  sigma.within~dunif(0.01, 10)
  mu.mex ~ dnorm(4, 0.001)
  tau.mex<-1/(sigma.mex*sigma.mex)
  sigma.mex~dunif(0.01, 10)
  mu.plac ~ dnorm(4, 0.001)
  tau.plac<-1/(sigma.plac*sigma.plac)
  sigma.plac~dunif(0.01, 10)
  diff <- mu.plac - mu.mex
}
```

In which:

Real data:

Stiff_Plac[i, t]	IVR stiffness score for patient i at time t during placebo treatment
Stiff_Mex[i, t]	IVR stiffness score for patient i at time t during mexiletine treatment
N_plac[i]	Number of observations for patient i during placebo treatment
N_mex[i]	Number of observations for patient i during mexiletine treatment
n	Number of patients

Estimated model parameters

<i>mu_mex[i]</i>	mean IVR score of patient i during the placebo treatment
<i>mu_plac[i]</i>	mean IVR score of patient i during the mexiletine treatment
<i>mu.plac</i>	Mean IVR score of NDM population (so the mean of the individual means) during placebo treatment (=population mean)
<i>sigma.plac</i>	Standard deviation of the individual mean IVR scores in the NDM population during placebo treatment
<i>tau.plac</i>	$1/(\text{sigma.plac} * \text{sigma.plac})$
<i>mu.mex</i>	Mean IVR score of NDM population (so the mean of the individual means) during mexiletine treatment (=population mean)
<i>sigma.mex</i>	Standard deviation of the individual mean IVR scores in the NDM population

	during placebo treatment
<i>tau.mex</i>	$1/(\sigma_{\text{mex}} \cdot \sigma_{\text{mex}})$
<i>sigma.within</i>	Standard deviation for IVR score measurements for an individual patient. It is assumed that the
	standard deviation is the same during placebo and mexiletine treatment.
<i>tau.within</i>	$1/(\sigma_{\text{within}} \cdot \sigma_{\text{within}})$
<i>diff</i>	$\mu_{\text{plac}} - \mu_{\text{mex}}$
population	Difference between population mean IVR score during placebo treatment and
	mean IVR score during mexiletine treatment (=population effect)
<i>p1</i>	Probability that the population effect will be greater than 0
<i>P2</i>	Probability that the population effect will be greater than 0.75
<i>P3</i>	Probability that the population effect will be greater than 2

eMethods 3. Winbugs Bayesian Hierarchical Model to Estimate the Mean Effect of Mexiletine in Genotype Subgroups

```
model
{
  for( i in 1 : n ) {
    mu_gen.mex[i] <- a + b*Gen_Pat[i]
    mu_gen.plac[i] <- c + d*Gen_Pat[i]
    mu_mex[i] ~ dnorm(mu_gen.mex[i], tau.mex)
    mu_plac[i] ~ dnorm(mu_gen.plac[i], tau.plac)
    diff_patient[i] <- mu_plac[i] - mu_mex[i]
    for ( t in 1 : N_plac[i] ) {
      Stiff_Plac[i, t] ~ dnorm(mu_plac[i], tau.within)
    }
    for ( t in 1 : N_mex[i] ) {
      Stiff_Mex[i, t] ~ dnorm(mu_mex[i], tau.within)
    }
  }
}
```

```
p2_SCN4A<-step(diff_SCN4A - 0.75)
p2_CLCN1<-step(diff_CLCN1 - 0.75)
p3_SCN4A<-step(diff_SCN4A - 2)
p3_CLCN1<-step(diff_CLCN1 - 2)
```

```
tau.within<-1/(sigma.within*sigma.within)
sigma.within~dunif(0.01, 10)
a ~ dnorm(0, 0.01)
b ~ dnorm(0, 0.01)
c ~ dnorm(0, 0.01)
d ~ dnorm(0, 0.01)
```

```
tau.mex<-1/(sigma.mex*sigma.mex)
sigma.mex~dunif(0.01, 10)
tau.plac<-1/(sigma.plac*sigma.plac)
sigma.plac~dunif(0.01, 10)
mu.mex_SCN4A <- a
mu.mex_CLCN1 <- a + b
mu.plac_SCN4A <- c
mu.plac_CLCN1 <- c + d
diff_SCN4A <- mu.plac_SCN4A - mu.mex_SCN4A
diff_CLCN1 <- mu.plac_CLCN1 - mu.mex_CLCN1
}
```

In which:

Real data:

Stiff_Plac[i, t]	IVR stiffness score for patient i at time t during placebo treatment
Stiff_Mex[i, t]	IVR stiffness score for patient i at time t during mexiletine treatment
N_plac[i]	Number of observations for patient i during placebo treatment
N_mex[i]	Number of observations for patient i during mexiletine treatment
n	Number of patients

Gen_Pat[i] Genotype for patient i (1=CLCN1, 0=SCN4A)

Estimated model parameters

mu_mex[i] mean IVR score of patient i during the placebo treatment

mu_plac[i] mean IVR score of patient i during the mexiletine treatment

mu_plac_SCN4A Mean IVR score of SCN4A population (so the mean of the individual means) during placebo treatment (=population mean)

mu_plac_CLCN1 Mean IVR score of CLCN1 population (so the mean of the individual means) during placebo treatment (=population mean)

sigma.plac Standard deviation of the individual mean IVR scores in the NDM population during placebo treatment (assumed to be the same for both genotypes)

tau.plac $1/(\text{sigma.plac} * \text{sigma.plac})$

mu_mex_SCN4A Mean IVR score of SCN4A population (so the mean of the individual means) during mexiletine treatment (=population mean)

mu_mex_CLCN1 Mean IVR score of CLCN1 population (so the mean of the individual means) during mexiletine treatment (=population mean)

sigma.mex Standard deviation of the individual mean IVR scores in the NDM population during placebo treatment (assumed to be the same for both genotypes)

tau.mex $1/(\text{sigma.mex} * \text{sigma.mex})$

sigma.within Standard deviation for IVR score measurements for an individual patient. It is assumed that the standard deviation is the same during placebo and mexiletine treatment and the same for both genotypes.

tau.within $1/(\text{sigma.within} * \text{sigma.within})$

diff_SCN4A $\text{mu.plac} - \text{mu.mex}$ for SCN4A patients

Difference between population mean IVR score during placebo treatment and mean IVR score during mexiletine treatment (=population effect) for SCN4A patients

diff_CLCN1 $\text{mu.plac} - \text{mu.mex}$ for CLCN1 patients

P2_SCN4A Probability that the population effect for SCN4A population will be greater than 0.75

P2_CLCN1 Probability that the population effect for CLCN1 population will be greater than 0.75

P3_SCN4A Probability that the population effect for SCN4A population will be greater than 2

P3_CLCN1 Probability that the population effect for CLCN1 population will be greater than 2

eMethods 4. Details on the Simulation-Based Sample Size Calculation

We performed two sets of simulations to determine the operating characteristics of the overall strategy (combined N-of-1 trials using the Bayesian hierarchical model meta-analysis) - one set under the assumption of no treatment effect (for the determination of the type 1 error), and one set under the assumption of a particular fixed treatment effect (for the determination of the power). As fixed treatment effect we used 1.75 point difference on the IVR-stiffness scale, that was based on preliminary results of the previous RCT by Statland et al. (2012), provided by the study team.

1.1 Calculation of type 1 error rate - simulations under the assumption of a treatment effect of zero

1. For each virtual patient, using SAS 9.4 software, we simulated data (14 observations per arm per treatment pair) that allowed the patient to have their own randomly chosen baseline value and expected variability, both for the placebo and mexiletine treatment periods (with the overall treatment effect being zero);
2. Per individual patient, the Bayesian interim analysis defined in eMethods 2 (using Winbugs version 1.4 software, run from R Statistics version 3.1.3) was performed to determine whether a conclusion of no meaningful effect (<20% posterior probability on 0.75 point difference) or meaningful effect (>80% posterior probability on 0.75 point difference) would be drawn, or the patient would be evaluated with another treatment pair (between 20-80% posterior probability on 0.75 point difference);
3. Step 1 and 2 were continued until no further testing on that patient would have been conducted, and the final results for that patient was recorded (mean treatment effect with 95% credible interval and posterior probability on a meaningful treatment effect);
4. Step 1-3 were repeated until 30 virtual patient data-sets were simulated (with variation in the number of treatment-sets). All data from these 30 virtual individual N-of-1 trials was used to perform the Bayesian meta-analysis on a group level according to eMethods 3;
5. We recorded the overall result (i.e., posterior probability on a meaningful treatment effect on a group level) of the Bayesian meta-analysis and the final estimate of the treatment effect (mean effect, 95% credible interval);
6. Step 1-5 were repeated 1000 times (corresponding with data from 1000 aggregated N-of-1 trials) and we determined: (a) the fraction of meta-analyses that return a positive conclusion (i.e. a >80% posterior probability on a meaningful treatment effect): as measure for the type

l error rate. The type 1 error rate (number of times that p2.mean.tot > 80%: (p2_effect=1)) was found to be 0.00 (see Table 2 below); (b) the mean treatment effect from these 1000 simulated data-sets, as measure for the bias since the true value is exactly zero, was found to be 0.07 (see Table 1 below).

		mean_sets	diff.mean.tot	mu.mex.tot	mu.plac.tot	p1.mean.tot	p2.mean.tot
N	Valid	1000	1000	1000	1000	1000	1000
	Missing	0	0	0	0	0	0
Mean		3.41	.07	4.58	4.65	.55	.12
Median		3.43	.06	4.57	4.66	.55	.09
Std. Deviation		.21	.27	.38	.31	.19	.10
Minimum		2.73	-.85	3.46	3.74	.04	.00
Maximum		3.97	.95	5.82	5.57	.97	.66
Percentiles	25	3.27	-.12	4.32	4.43	.41	.04
	50	3.43	.06	4.57	4.66	.55	.09
	75	3.57	.25	4.83	4.86	.69	.17

Table 1. Results from simulations under the assumption of a treatment effect of zero

Mean_sets = mean number of pairs per patient

Diff.mean.tot = mean treatment effect (mu.plac.tot – mu.mex.tot)

Mu.mex.tot = mean effect during mexiletine period

Mu.plac.tot = mean effect during placebo period

P1.mean.tot = probability of an effect > .00

P2.mean.tot = probability of an effect > .75

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	.00	1000	100	100	100
	1.00	.00	.00	.00	100
	Total	1000	100	100	100

Table 2. Results from calculation of type 1 error rate

Type 1 error rate = number of times that p2.mean.tot > 80%: (p2_effect=1)

1.2 Calculation of the power - simulations under the assumption of a treatment effect of 1.75

(Step 1 – 5) Calculation the power was analogous to the above described steps 1-6 for calculation of the type 1 error with the exemption that the overall treatment effect for each virtual patient within each virtual treatment-set was kept at a fixed treatment effect of 1.75 (instead of zero).

- Step 1-5 were repeated 1000 times (corresponding with data from 1000 aggregated N-of-1 trials) to determine: (a) the fraction of meta-analyses that return a positive conclusion (i.e. a >80% posterior probability on a meaningful treatment effect): as measure for the power. The power (number of times that p2.mean.tot > 80%: (p2_effect=1)) was found to be 68.5% (see

Table 4 below); (b) the mean treatment effect from these 1000 simulated data-sets (1.29 – see Table 3 below) minus the assumed, fixed treatment effect of 1.75 (as measure for the bias) was found to be -0.46 [1.29 – 1.75].

		mean_sets	diff.mean.tot	mu.mex.tot	mu.plac.tot	p1.mean.tot	p2.mean.tot
N	Valid	1000	1000	1000	1000	1000	1000
	Missing	0	0	0	0	0	0
Mean		2.59	1.29	3.38	4.67	.99	.83
Median		2.60	1.29	3.37	4.67	1.00	.86
Std. Deviation		.25	.26	.35	.31	.02	.12
Minimum		1.73	.43	2.44	3.79	.81	.26
Maximum		3.47	2.13	4.60	5.59	1.00	1.00
Percentiles	25	2.43	1.12	3.13	4.46	.99	.77
	50	2.60	1.29	3.37	4.67	1.00	.86
	75	2.73	1.46	3.62	4.88	1.00	.93

Table 3. Results from simulations under the assumption of a treatment effect of 1.75

Mean_sets = mean number of pairs per patient.

Diff.mean.tot = mean treatment effect (mu.plac.tot – mu.mex.tot)

Mu.mex.tot = mean effect during mexiletine period

Mu.plac.tot = mean effect during placebo period

P1.mean.tot = probability of an effect > .00

P2.mean.tot = probability of an effect > .75

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	.00	315	31,5	31.5	31.5
	1.00	685	68.5	68.5	100.0
	Total	1000	100.0	100.0	

Table 4. Results from power calculation

Power = number of times that p2.mean.tot > 80%: (p2_effect=1)